

Experiment 1: Calibration of Volumetric Glassware

In this laboratory experiment, the three types of glassware typically used by an analytical chemist: a volumetric flask, volumetric pipets, and a measuring cylinder will be calibrated using the density data of water at room temperature. These tools are used extensively when performing gravimetric and titrimetric Analyses. To avoid introducing systematic errors in measurements, each of these instruments must be properly calibrated. And, to reduce the random errors inherent when using these instruments, their proper use must be thoroughly understood. The quality of the measurements obtained from these tools depends heavily on the care taken in calibrating and in using each instrument.

Tips for Correct Use of Volumetric Glassware Pipettes

The pipette is used to transfer a volume of solution from one container to another. Most volumetric pipettes are calibrated To-Deliver (TD) with a certain amount of the liquid remaining in the tip and as a film along the inner barrel after delivery of the liquid. The liquid in the tip should not be blown out.

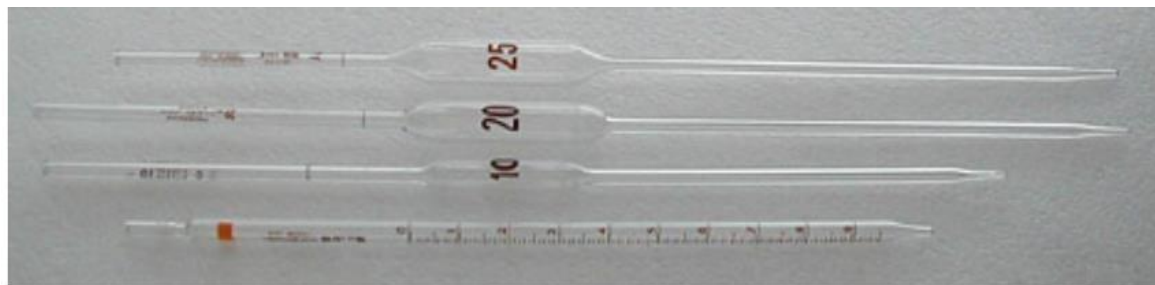


Figure 1. Volumetric and Measuring Pipettes.

To fill the pipet, insert it vertically in the liquid, with the tip near the bottom of the container. Apply suction to draw the liquid above the graduation mark. Quickly place a forefinger over the end of the stem. Withdraw the pipet from the liquid and use a dry paper to wipe off the stem. Now place the tip of the pipet against the container from which the liquid has been withdrawn and drain the excess liquid such that the meniscus is at the graduation mark. Move the pipet to the receiving container and allow the liquid to flow out (avoiding splashing) of the pipet freely. When most of the liquid has drained from the pipet, touch the tip to the wall of the container until the flow stops and for an additional count of ten.

Volumetric Flasks

The volumetric flask is used to prepare Standard Solutions or in diluting a sample. Most of these flasks are calibrated To-Contain (TC) a given volume of liquid. When using a flask, the solution or solid to be diluted is added and solvent is added until the flask is about two-thirds full. It is important to rinse down any solid or liquid that has adhered to the neck. Swirl the solution until it is thoroughly mixed. Now add solvent until the meniscus is at the calibration mark. If any droplets of solvent adhere to the neck, use a piece of tissue to blot these out. Stopper the flask securely and invert the flask at least ten times.



A General Calibration Procedure

As was noted above, volumetric glassware is calibrated by measuring the mass of water that is contained in or delivered by the device. This mass data is then converted to volume data using the tabulated density of water (See Appendix) at the temperature of calibration. Finally, this volume data is corrected to the standard temperature of 20 °C or the room temperature. In this experiment, a volumetric flask, a measuring pipette, and a volumetric pipette will be using gravimetrically determined water. In each case, the measured mass of the calibrating water will be standardized to 20 °C or room temperature. The density of water at different

temperatures is given below.

Temperature (°C)	Density (g/mL)
15	0.999 102 6
16	0.998 946 0
17	0.998 777 9
18	0.998 598 6
19	0.998 408 2
20	0.998 207 1
21	0.997 995 5
22	0.997 773 5
23	0.997 541 5
24	0.997 299 5
25	0.997 047 9
26	0.996 786 7
27	0.996 516 2
28	0.996 236 5
29	0.995 947 8
30	0.995 650 2

1. Take 3 different volumes of water each from a pipette, flask, and measuring cylinder and measure the weight of the liquid.
2. Note the difference between the actual weight and the calculated one and estimate the error up to 3 decimal places.

All the results should be given in the tabulated form.